

Problem 6

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Problem 1 (a) Find the following definite integral:

$$\int_{-4}^4 \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}} dx$$

Explanation. Let $f(x) = \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}}$. Then notice that

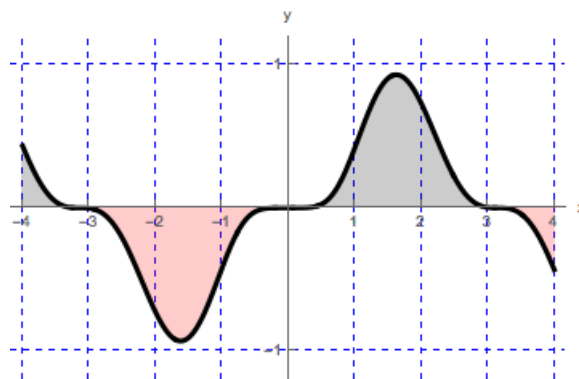
$$\begin{aligned} f(-x) &= \frac{(-x)^2 \sin^3(-x)}{\sqrt{(-x)^4 + 1}} \\ &= \frac{x^2 (-\sin(x))^3}{\sqrt{x^4 + 1}} \quad (\text{since } \sin(x) \text{ is an odd function}) \\ &= \frac{-x^2 \sin^3(x)}{\sqrt{x^4 + 1}} \\ &= -f(x). \end{aligned}$$

Thus, f is an odd function and therefore

$$\int_{-4}^4 \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}} dx = 0.$$

We can illustrate our computation with the graph of the function f , where

$$f(x) = \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}}, \text{ for } -4 \leq x \leq 4.$$



- (b) Suppose that f is an even function. Given that $\int_0^6 f(x) dx = 13$, find $\int_{-6}^6 (5f(x) + 14) dx$.

Explanation. First notice that

$$\int_{-6}^6 (5f(x) + 14) dx = 5 \int_{-6}^6 f(x) dx + \int_{-6}^6 14 dx. \quad (1)$$

Since f is even, we know that

$$\int_{-6}^6 f(x) dx = 2 \int_0^6 f(x) dx = 2(13) = 26. \quad (2)$$

We also know that $g(x) = 14$ is also an even function. We have: $\int_{-6}^6 14 dx =$
 $2 \int_0^6 14 dx$

$$2 \int_0^6 14 dx = 2(14)(6) = 168. \quad (3)$$

Then substituting equations (2) and (3) into equation (1) gives:

$$\int_{-6}^6 (5f(x) + 14) dx = 5(26) + 168 = 130 + 168 = 298.$$